

Unit 4 – Additional Review Exercises

1. Find two positive numbers whose sum is 20 such that their product is as large as possible.
2. One number is twelve more than another. What numbers give the least product?
3. What positive number plus its reciprocal gives the least possible sum?
4. Two numbers add to 12. Find them if twice one of them plus their product is a maximum.
5. Separate 120 into two parts such that the product of one part and the square of the other is a maximum.
6. Find two positive numbers whose product is 20 and the sum of whose squares is a minimum.
7. What number exceeds its positive square root by the least amount?
8. What positive number exceeds its cube by the maximum amount?
9. Find two numbers whose sum is 10 such that twice the product of the cube of the first and the square of the second is a maximum.
10. Find the area of the largest rectangle having a perimeter of 200 m.
11. What are the dimensions of the rectangle of maximum area if its perimeter is 60 m?
12. Find the dimensions of the rectangle of greatest area that can be inscribed in a semicircle of radius 10 cm.
13. Find two numbers whose sum is 32 such that the sum of their squares is a minimum.
14. What is the minimum perimeter a rectangle can have and yet contain an area of 100 cm^2 ?
15. A farmer wishes to fence in a rectangular region using 200 m of fence. He will use the side of his barn as one side of the fence. If the area of the region is to be a maximum, find the dimensions of the region.
16. Find the area of the largest isosceles triangle having a perimeter of 18 cm.
17. A square piece of tin, 24 cm by 24 cm, is to be made into an open box by cutting out equal squares from the corners and bending up the flaps. Find the length of the side of the cutout square so that the volume of the box shall be a maximum.
18. Find the length of the base of an isosceles triangle of maximum area if its two equal legs are 12 cm long.

19. A rectangular box with a square base and an open top is to be constructed. Find the dimensions of the largest box that can be made from 675 cm^2 of material.
20. Find the volume of the largest cylindrical container that can be inscribed in a sphere of radius r .

Answers:

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| 1. 10,10 | 2. $-6,6$ | 3. 1 | 4. 7,5 |
| 5. 80,40 | 6. $2\sqrt{5}, 2\sqrt{5}$ | 7. $\frac{1}{4}$ | 8. $\frac{\sqrt{3}}{3}$ |
| 9. 6,4 | 10. 2500 m^2 | 11. 15m x 15m | 12. $5\sqrt{2} \times 10\sqrt{2} \text{ cm}$ |
| 13. 16,16 | 14. 40 cm | 15. 50 x 100 m | 16. $9\sqrt{3} \text{ cm}^2$ |
| 17. 4 x 4 cm | 18. $12\sqrt{2} \text{ cm}$ | 19. 15 x 15 x 7.5 cm | 20. $\frac{4\pi r^3}{3\sqrt{3}}$ |